# Benedikt Safigan

benedikt.safigan@colorado.edu | (720) 232-3787 | Boulder, CO | LinkedIn | Portfolio | GitHub

## PROFESSIONAL SUMMARY

Computer Science student at CU Boulder with hands-on, full-stack internship experience, from low-level embedded systems to React/Node.js web applications.

## PROFESSIONAL EXPERIENCE

**Software Engineer (Contractor) | GovWorx** Jun 2026 – Present

- Designed and shipped **three** CAD (Computer-Aided Dispatch) connectors from scratch – Mark43 REST API, IMC over Actian Zen ODBC, and CentralSquare OneSolution XML – for GovWorxDataAgent, an on-premises .NET service streaming real-time 911 emergency data to a cloud AI platform; owned schema mapping, query logic, and configuration for each.
- Extended and hardened production connectors for public-safety vendors (Tyler, Versaterm, Hexagon, Spillman) across **6** data-source types – SQL Server, Oracle, Informix, ODBC, REST, and XML.
- Shipped **50+** Linear-tracked issues across **170+** commits via a reviewed pull-request workflow, covering backfill reliability, timezone-correctness fixes, per-connector telemetry, and production fixes under pressure – including cursor preservation on failed pulls to prevent data loss.
- Built a full-stack connectivity and firewall-diagnostics tool with a React/TypeScript frontend and an ASP.NET Core API, instrumented with OpenTelemetry and Datadog metrics.

**Firmware Engineering Intern | Exact Assembly** May 2023 – Aug 2023

- Developed and debugged **1000+** lines of embedded firmware for control systems at a tech contracting startup.
- Implemented and tuned PID control loops to achieve stable, responsive closed-loop behavior on real hardware.
- Collaborated with the engineering team to integrate, test, and validate firmware against system requirements.

**Programmer & Field Technician | VARSITY FIRST Robotics Team 2996** Aug 2021 – May 2024

- Led software development for a team of 5, delivering fully functional competition robots within **6-week build cycles** for **3 consecutive** seasons, achieving **100% on-time delivery**.
- Wrote **2,000+ lines** of autonomous and teleoperated robot control code in Java using WPILib, achieving **95%+** complete autonomous routine success rate across **6 regional competitions with zero forfeits**.

## PROJECTS

**BlitzSort** | C++17, Performance Optimization, CPU Cache, Algorithms

Cache-conscious quicksort achieving **128x** speedup on worst-case (reverse-sorted) data via median-of-three pivot selection, a 16-element insertion-sort threshold, and tail-recursion optimization – while holding O(n log n) and matching std::sort byte-for-byte.

**order-me-this** | C++17, Data Structures, Performance Optimization, Systems Design

A minimal C++17 limit order book benchmarking three matching-engine data structures (std::map, unordered_map, sorted deque); isolating the matching hot path revealed a **3.7x** latency win for sorted deque.

**Accelerated Image Processing** | OpenCL, C++17, GPU Computing, Parallel Programming

GPU-accelerated image-filter pipeline (grayscale, Gaussian blur, Sobel edge detection) in OpenCL, benchmarked against a scalar CPU baseline to achieve **15–26x** speedups on compute-bound filters.

**Clockless Mutex** | C++17, Distributed Systems, Concurrency

From-scratch distributed mutual-exclusion system implementing Lamport logical clocks over an unreliable message-passing network, using chaos injection to prove deadlock-free, causally-ordered operation without a shared clock.

## EDUCATION

**University of Colorado Boulder** | B.S. Computer Science Sept 2024 – May 2028

Minor: Computer Engineering | GPA: 3.44

## SKILLS

**Languages:** C, C++17, C#, Python, Bash, SQL, Java, JavaScript/TypeScript

**Systems & Performance:** Embedded Firmware, Control Systems & PID, Concurrency & Parallel Programming, Distributed Systems, Performance Optimization, CPU Cache & Memory Hierarchy, GPU Computing (OpenCL)

**Software & Tools:** Data Structures & Algorithms, REST APIs, .NET / ASP.NET Core, Git, GDB, Linux/Unix, Docker